



Available online at www.sciencedirect.com

ScienceDirect



RESEARCH ARTICLE

In silico genome-wide identification, phylogeny and expression analysis of the *R2R3-MYB* gene family in *Medicago truncatula*

ZHENG Xing-wei, YI Deng-xia, SHAO Lin-hui, LI Cong



Institute of Animal Sciences, Chinese Academy of Agricultural Sciences, Beijing 100193, P.R.China

Abstract

The *R2R3-MYB* genes make up one of the largest transcription factor families in plants, and play regulatory roles in various biological processes such as development, metabolism and defense response. Although genome-wide analyses of this gene family have been conducted in several species, *R2R3-MYB* genes have not been systematically analyzed in *Medicago truncatula*, a sequenced model legume plant. Here, we performed a comprehensive, genome-wide computational analysis of the structural characteristics, phylogeny, functions and expression patterns of *M. truncatula* *R2R3-MYB* genes. DNA binding domains are highly conserved among the 155 putative MtR2R3-MYB proteins that we identified. Chromosomal location analysis revealed that these genes were distributed across all eight chromosomes. Results showed that the expansion of the MtR2R3-MYB family was mainly attributable to segmental duplication and tandem duplication. A comprehensive classification was performed based on phylogenetic analysis of the *R2R3-MYB* gene families in *M. truncatula*, *Arabidopsis thaliana* and other plant species. Evolutionary relationships within clades were supported by clade-specific conserved motifs outside the MYB domain. Species-specific clades have been gained or lost during evolution, resulting in functional divergence. Also, tissue-specific expression patterns were investigated. The functions of stress response-related clades were further verified by the changes in transcript levels of representative *R2R3-MYB* genes upon treatment with abiotic and biotic stresses. This study is the first report on identification and characterization of *R2R3-MYB* gene family based on the genome of *M. truncatula*, and will facilitate functional analysis of this gene family in the future.

Keywords: *R2R3-MYB*, *Medicago truncatula*, gene family, stress response, function prediction

1. Introduction

Transcription factors (TFs) play significant roles in the regulation of gene expression by activating or inhibiting the tran-

scription of target genes (Mizoi *et al.* 2012). Most known TFs are grouped into different families based on their DNA-binding domains and structures (Riechmann *et al.* 2000).

MYB proteins constitute one of the best-characterized TF families and they are present in all eukaryotes (Dubos *et al.* 2010; Feng *et al.* 2015; Li *et al.* 2016). The MYB superfamily is defined by the presence of a highly conserved DNA-binding domain near the N-terminus (Stracke *et al.* 2014). This domain typically contains one to four imperfect repeats (R1, R2, and R3), each of which is approximately 50–53 amino acid (aa) residues in length and encodes three α -helices (Li *et al.* 2015). Within each repeat, the second and third helices form a helix-turn-helix (HTH) structure that extends

Received 3 June, 2016 Accepted 9 November, 2016
ZHENG Xing-wei, E-mail: smilezxw@126.com; Correspondence
LI Cong, Tel: +86-10-62815998, E-mail: licong0520@sina.com

© 2017 CAAS. Publishing services by Elsevier B.V. All rights reserved.
doi: 10.1016/S2095-3119(16)61521-6

into the major groove of DNA and mediates recognition of the target sequence (Martin and Paz-Ares 1997; Li *et al.* 2016). In contrast, the highly divergent C-terminus of MYB proteins contains the transcription activation/repression domain, which confers the regulatory role for each family member (Kranz *et al.* 1998; Franco-Zorrilla *et al.* 2014).

MYB proteins can be divided into different groups based on the number of adjacent repeats: 2R (R2R3-MYB), 3R (R1R2R3-MYB), 4R (R1R2R2R1/2-MYB) and 1R-MYB (MYB-related proteins) (Hou *et al.* 2014; Du *et al.* 2015; Liu *et al.* 2015). In plants, R2R3-MYBs are the most abundant type, making up one of the largest transcription factor families and are the most similar to their vertebrate homologs, the c-MYBs (Geethalakshmi *et al.* 2015). The R2R3-MYB genes probably evolved from an R1R2R3-MYB gene ancestor through loss of the R1 repeat (Lipsick 1996; Braun and Grotewold 1999) or from an R1-MYB gene through an intragenic domain duplication of the R1 repeat (Jiang *et al.* 2004).

The first cloned plant MYB gene *COLORED1* (C1) was isolated from maize (*Zea mays*), and encodes a R2R3-MYB transcription factor involved in anthocyanin biosynthesis (Paz-Ares *et al.* 1987). Subsequently, an increasing number of R2R3-MYB proteins were characterized by genetic approaches and shown to regulate a number of biological processes, including cell shape and identity (Perez-Rodriguez *et al.* 2005; Baumann *et al.* 2007; Lau *et al.* 2015; Kim *et al.* 2016), developmental control (Muller *et al.* 2006; Song *et al.* 2011; Ambawat *et al.* 2013; Xu *et al.* 2016; Zheng *et al.* 2016), hormone signal transduction (Gocal *et al.* 1999; Abe *et al.* 2003; Yu *et al.* 2016), response to abiotic stresses and pathogen defense (Zhang *et al.* 2012; Bonthala *et al.* 2016; Liu *et al.* 2016; Wang H Y *et al.* 2016). In recent years, accumulating evidence has confirmed that numerous R2R3-MYBs are key factors in regulatory networks controlling primary and secondary metabolism. R2R3-MYB subfamily members in combination with bHLH TFs and WD40 proteins form a ternary MYB-bHLH-WD40 complex to regulate the expression of structural genes controlling the late steps of the anthocyanin and proanthocyanidins (condensed tannins) biosynthetic pathway (Albert 2015; Passeri *et al.* 2016; Perez-Diaz *et al.* 2016). Additionally, R2R3-MYB TFs are involved in the regulation of secondary cell wall formation, by modulating the transcription of lignin, xylose and cellulose biosynthetic genes (Chai *et al.* 2014; Wang W *et al.* 2016). Moreover, a newly identified R2R3-MYB protein, RCP1, positively regulates carotenoid biosynthesis during flower development (Sagawa *et al.* 2016).

Due to its diploidy and considerably small genome (haploid size 500–550 Mb), ability of nitrogen fixation, high genetic transformation efficiency, and rapid life cycle with high seed yield, *Medicago truncatula* is considered a model

species to analyze biological processes that are unique or pertinent to legumes (Young *et al.* 2011). Since the release of several whole genome sequences, R2R3-MYB genes have been well annotated in *Arabidopsis thaliana* (126 members) (Stracke *et al.* 2001), *Oryza sativa* (88 members) (Katiyar *et al.* 2012), *Z. mays* (157 members) (Du *et al.* 2012a), *Glycine max* (soybean, 244 members) (Du *et al.* 2012b), *Populus trichocarpa* (192 members) (Wilkins *et al.* 2009) and *Vitis vinifera* (108 members) (Matus *et al.* 2008). Nevertheless, a systematic analysis of the R2R3-MYB gene family in *M. truncatula* is still lacking. It is therefore of interest to conduct a systematical analysis of *M. truncatula* R2R3-MYB genes, including gene classification, chromosomal locations, phylogenetic relationships, conserved motifs as well as expression patterns.

2. Materials and methods

2.1. Database search for R2R3-MYB protein coding genes in the *M. truncatula* genome

The sequences of 126 *A. thaliana* R2R3-MYB proteins were downloaded from The Arabidopsis Information Resource (TAIR, www.arabidopsis.org) (Berardini *et al.* 2015). These sequences were used as queries in a BLAST search against the *M. truncatula* proteome sequences provided by The J. Craig Venter Institute (JCVI) *M. truncatula* Sequencing Project v4.0 (www.jcvi.org/medicago/) with an E-value cutoff of 0.001. The obtained MYB sequences were confirmed based on the presence of two or more intact MYB domains using the Pfam Program (<http://pfam.xfam.org/>) (Finn *et al.* 2016) and SMART database with an E-value cutoff of 1E-10 (<http://smart.embl-heidelberg.de/>) (Finn *et al.* 2016). Information about coding sequences (CDS), full-length gene sequences and intron/exon distribution patterns were also downloaded from the *M. truncatula* genome website. We obtained information on the chromosome locations of the putative R2R3-MYB genes from the results of BLASTP searches in the JCVI *M. truncatula* Genome Browser (JBrowse 1.11.1).

2.2. Sequence conservation analysis

Multiple sequence alignments of the 155 identified R2R3-MYB proteins in *M. truncatula* were conducted using MUSCLE with default parameters. To analyze the sequence features of the MYB domains of these proteins, the conserved region of each alignment was trimmed with Gblocks v0.91b, and BioEdit v7.0.5 was used to highlight conserved amino acids. The sequence logos for R2 and R3 repeats were generated using the Weblogo v3 Online Program (<http://weblogo.threeplusone.com/>).

The motifs conserved among MtR2R3-MYB members

were identified using the MEME v4.11.1 Online Program (<http://meme-suite.org/index.html>). The optimized parameters employed in the analysis were: Maximum number of motifs was 80, and the width of each motif was set from 6 to 250 residues (Bailey and Elkan 1994). Only motifs with an E-value <1E-20 were retained for further analysis.

2.3. Phylogenetic analyses and classification of *MtR2R3-MYB* gene family members

A phylogenetic analysis was performed to clarify the evolutionary relationship of the *MtR2R3-MYB* family to the *A. thaliana* R2R3-MYB family. An additional 86 functionally characterized R2R3-MYB TFs from other species were included in the analysis, with sequences downloaded from the Plant Transcription Factor Database (<http://planttfdb.cbi.pku.edu.cn/>). Members of the 3R and 4R-MYBs were also included in the analysis. Multiple alignments of full length sequences were performed using ClustalX2 with default parameters. The neighbor-joining (NJ) method was used to construct a phylogenetic tree using MEGA v6.06 with Poisson correction and 1000 bootstrap replicates (Tamura et al. 2011; Hall 2013). Classification of the *MtR2R3-MYB* genes was performed according to the phylogenetic relationship with their orthologs in *A. thaliana* and other landmark MYB genes.

2.4. *In silico* expression analysis using microarray data

The microarray datasets used to determine the expression profiles of the putative *R2R3-MYB* genes were retrieved from the *Medicago truncatula* Gene Expression Atlas (MtGEA, <http://mtgea.noble.org/v2/>), using the *R2R3-MYB* gene IDs as the queries. In addition, to study the expression patterns of *R2R3-MYB* genes in response to biotic and abiotic stresses, datasets profiling changes in gene expression in response to salt stress at six time points and in response to *Phymatotrichum* infection at three time points in the root were used (Uppalapati et al. 2008). The expression profiles of the *MtR2R3-MYB* genes were obtained using the program Heml ver. 1.0.3.3 (Deng et al. 2014) with an average linkage hierarchical clustering algorithm, and visualized in heat maps generated using the same software.

2.5. Plant materials and stress treatments

Seeds of *M. truncatula* (cv. Jemalong A17) were scarified and germinated on Petri plates for 3 days in the dark. Young seedlings were then transplanted into 1/2-strength Hoagland solution (Zhang et al. 2014) and grown hydroponically in a growth chamber maintained at 25/22°C (day/night), with a

16-h light/8-h dark photoperiod and 75% relative humidity.

Three-week-old seedlings were subjected to several stress treatments. For salt and drought stress, the seedlings were treated with 200 mmol L⁻¹ NaCl and 300 mmol L⁻¹ mannitol solution, respectively, for 2, 5, 10, and 24 h. For the cold stress treatment, the seedlings were incubated at 4°C for 2, 6, 12, and 24 h. For the ABA treatment, the seedlings were sprayed with 100 μmol L⁻¹ ABA solution and leaves were harvested after 2, 6, 12, and 24 h. The roots from seedlings that received salt and drought stress treatments, and the leaves from seedlings that received cold stress and ABA treatments were harvested separately, immediately frozen in liquid nitrogen, and kept at -80°C until use.

2.6. Quantitative real-time PCR analysis

Total RNA was isolated with TRIzol® reagent (Invitrogen, Carlsbad, CA, USA) and quantified spectrophotometrically. First-strand cDNA was synthesized using the FastQuant RT Kit (with gDNase) (Tiangen, Beijing, China) in a 20-μL reaction mixture containing 1 μg total RNA. A four-fold dilution of the cDNA was used as the template for PCR. qRT-PCR was carried out with a 7500 Real-Time PCR System (Applied Biosystems, Foster City, CA, USA) using SYBR® Premix Ex Taq® II (Tli RNaseH Plus) (TaKaRa Bio., Dalian, China). The PCR conditions were as follows: 30 s at 95°C, followed by 40 cycles of 95°C for 5 s and 60°C for 34 s. The experiments were assigned three biological replicates. The relative expression level of each gene was calculated using the $\Delta\Delta C_t$ method and was normalized to the *MtActin* gene (GenBank ID: JQ028731.1) (Livak and Schmittgen 2001). The mean C_t value for each sample and standard errors of the mean among replicates were calculated with ABI 7500 software, ver. 2.0.6 (Life Technologies, Carlsbad, CA, USA). Data are summarized in Appendix A, and the primers used are listed in Appendix B.

3. Results

3.1. Identification of 155 genes encoding putative R2R3-MYB TFs in the *M. truncatula* genome

To identify R2R3-MYB encoding genes in the *M. truncatula* genome, a BLASTP search was performed using known *A. thaliana* R2R3-MYB protein sequences as queries. In order to verify the identity of these MYB sequences, Pfam and SMART searches were then used to ensure that they contained two or more MYB repeats. In addition, these putative *R2R3-MYB* genes were also inspected to make sure they possess complete open reading frames and map to unique genomic loci. A total of 155 non-redundant *R2R3-MYB* sequences were obtained and named *MtMYB0*

to *MtMYB154*, based on the order of their chromosome locations in the *M. truncatula* genome browser (Appendix C). In addition, seven proteins with multiple MYB repeats that are distantly related to the typical R2R3-MYB proteins were identified: five R1R2R3-MYB proteins (3R-MYB) and two 4R-MYB proteins (Appendix C). Detailed information about the *MtMYBs*, including gene name, locus ID, genomic position, intron number, protein length, molecular weight and isoelectric point is listed in Appendix C. The 155 predicted R2R3-MYB proteins range from 175 (*MtMYB71*) to 1513 aa (*MtMYB114*) in length, with an average length of 334.21 aa (Appendix C).

The exonic and intronic regions were determined for each *MtR2R3-MYB* gene by comparing the coding sequence to the gene sequence. The majority of the *MtR2R3-MYB* genes (122) contain two introns, 18 *MtR2R3-MYBs* possess one intron and six *MtR2R3-MYBs* have three introns. The analysis also revealed five genes with no introns, whereas two genes with nine introns, and *MtMYB19* and *MtMYB86* contained 10 and 11 introns, respectively (Appendix C).

3.2. Analysis of the conserved MYB domains in *M. truncatula* R2R3-MYB TFs

To investigate the sequence features of R2R3-MYB proteins in *M. truncatula*, multiple sequence alignments were performed with MUSCLE followed by trimming with Gblocks, leading to the identification of a highly conserved 108 aa region among all R2R3-MYB proteins. This 108-aa region covers the R2R3-MYB domain, which is the hallmark of

the R2R3-MYB family (Appendix D). Consistent with other plant species (Wilkins *et al.* 2009; Soler *et al.* 2015), the R2 and R3 MYB repeats of the *MtR2R3-MYBs* contain characteristic amino acids, including six evenly spaced and highly conserved Trp (W) residues that play a key role in sequence-specific binding of DNA (Fig. 1). As shown in Fig. 1 and Appendix D, in most *MtR2R3-MYBs*, the first Trp residue (W_{62}) in the R3 repeat has been replaced by Phe (F), Ile (I), Leu (L), Met (M), or Tyr (Y). We also found that position 100 of the MYB repeat, which contains the third conserved tryptophan residue (W_{100}) in the R3 repeat, was not completely conserved in all of the *MtR2R3-MYBs* (Fig. 1, Appendix D). An Ala (A) and a Phe (F) residues, were found in *MtMYB86* and *MtMYB137*, respectively, both of which were encoded by alternatively spliced genes (*Medtr5g038910* and *Medtr7g115550*, Appendix C).

3.3. Chromosomal location and gene duplication events among the R2R3-MYB genes

Of the 155 *R2R3-MYB* genes, 145 are distributed unevenly among the eight *M. truncatula* chromosomes, and the remaining 10 genes are located on eight scaffolds that have not yet been anchored to any chromosome (Fig. 2, Appendix C). Chromosomes V, IV and VII contain relatively more *MtR2R3-MYB* genes, with 29, 24 and 23 genes, respectively. Chromosomes VI, II and VIII contain relatively few *MtR2R3-MYB* genes, with only 8, 13 and 14 genes, respectively. In addition, the *MtR2R3-MYB* gene density along each chromosome is unequal, with relatively higher densities of

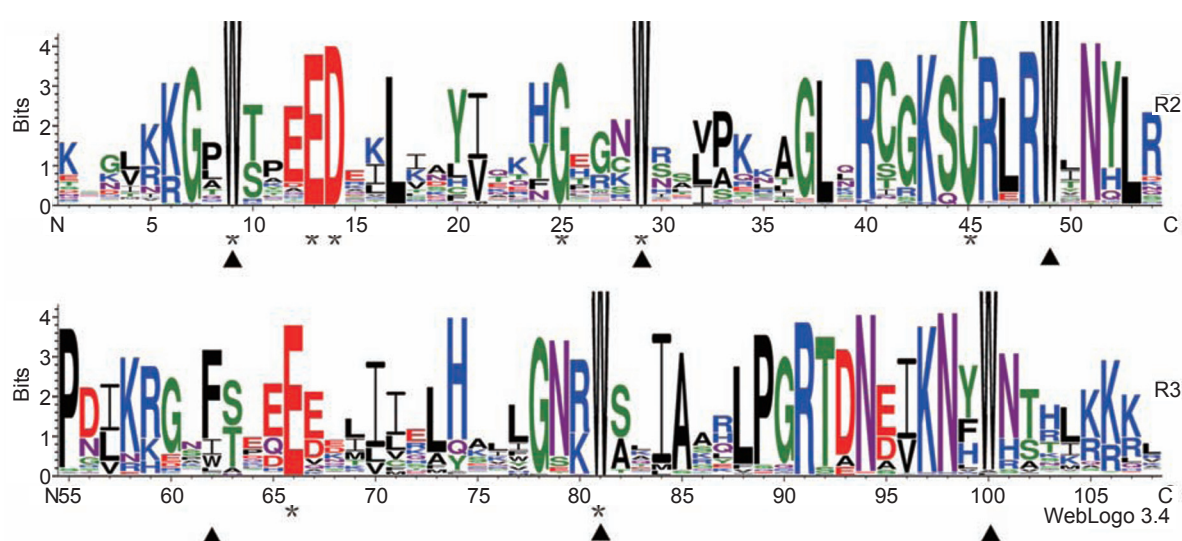


Fig. 1 The R2 and R3 MYB repeats are highly conserved across all R2R3-MYB proteins in *Medicago truncatula*. The sequence logos are based on full-length alignments of all *MtR2R3-MYB* domains. The bit score indicates the information content of each amino acid at that position in the sequence. * denotes the conserved residues that are identical among all 155 *MtR2R3-MYBs*, and ▲ denotes the highly conserved Trp residues in the MYB domain.

MtR2R3-MYB genes on the bottom of chromosomes I–V and on the top of chromosomes VI and VIII. On average, there is one *R2R3-MYB* gene every 2.65 Mb.

In plants, the generation of new gene family members is often the result of gene duplication (Sharoni *et al.* 2011). In order to elucidate the potential expansion mechanism of the *R2R3-MYB* gene family in *M. truncatula*, we identified both segmental and tandem duplication events. Gene duplicates were defined based on the following criteria: the length of the aligned sequence covers >70% of the longer gene, and the similarity of the aligned regions is >70% (Song and Nan 2014; He *et al.* 2016). Duplicated genes that adjacent to each other in a given chromosome, with no more than one intervening gene were defined as tandem duplicates, while duplicated genes on different chromosomes or within the same chromosome but not one following the other were defined as segmental duplicates (He *et al.* 2016). According to a whole genome analysis of gene duplication, 72 (approximately 45%) *MtR2R3-MYB* duplicated genes were identified, and of these 30 genes were produced by tandem duplication (Fig. 2). Tandem repeats most commonly include duplicate or triplicate *R2R3-MYB* genes (Wilkins *et al.* 2009), but the MtMYB104–109 cluster on chromosome V and the MtMYB142–147 cluster on chromosome VIII are two instances of tandem arrays with six genes. We found that 21 pairs are involved in segmental duplications (Fig. 2, linked with red lines). Among the segmental duplicated gene pairs, 16

are located in conserved positions on 11 duplicated blocks (Fig. 2, yellow bars with numbers), suggesting that they may have evolved from whole genome duplication events. The other five duplicate pairs are located on duplicated segments that are not part of larger collinear duplicated blocks (Fig. 2). These observations suggest that diverse duplication events have contributed to the expansion of *R2R3-MYB* gene family in *M. truncatula*.

3.4. Phylogenetic analysis and *de novo* prediction of motifs in MYB family genes

In order to investigate the phylogenetic relationships between *MtR2R3-MYB* genes and their homologs from other species, we constructed an unrooted phylogenetic tree based on the alignment of the amino acid sequences of 380 *R2R3-MYB* genes, including 162 from *M. truncatula* (*R3-* and *R4-MYB* genes included), 132 from *A. thaliana* and 86 well-characterized landmark genes from other 23 plant species (Appendix E). Based on the topology and robustness of the neighbor-joining (NJ) phylogenetic tree results, we resolved these *R2R3-MYB* genes into 46 clades (C1–C46), which ranged in size from two (C18) to 24 (C1). The distribution of the clades can be found in Appendix F. In our classification of the *MYB* genes, we also took into account the subgroup categories from *A. thaliana* (Stracke *et al.* 2001; Dubos *et al.* 2010). Generally, a good agreement existed between the phylogeny

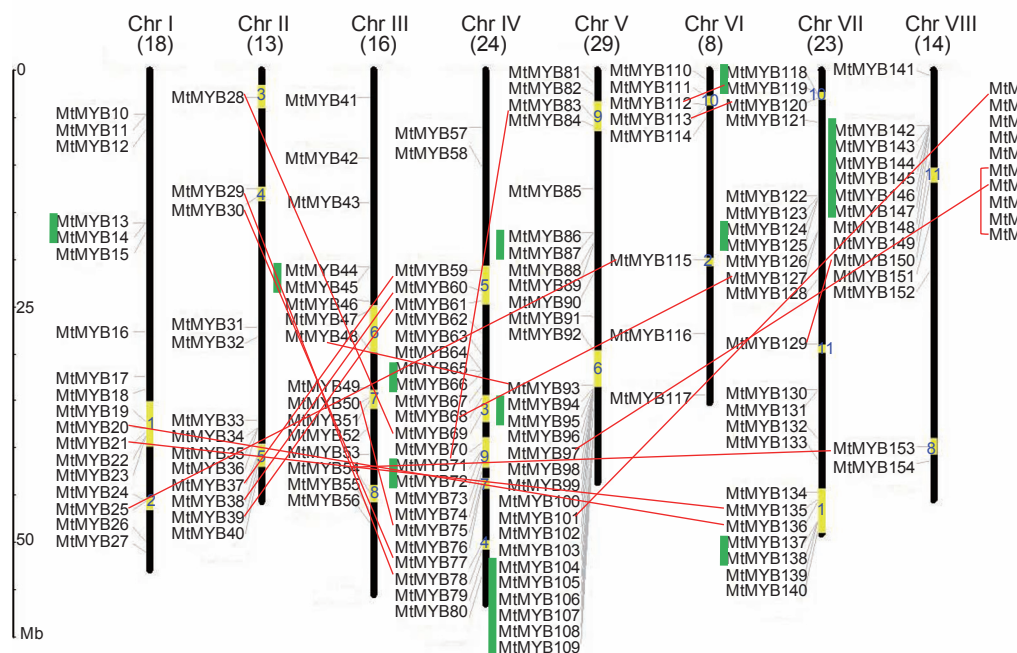


Fig. 2 Positions of *R2R3-MYB* family genes on the *Medicago truncatula* chromosomes. The chromosomal location of each *R2R3-MYB* was mapped according to the *M. truncatula* genome annotation. The chromosome name and the number of *MtR2R3-MYB* genes on that chromosome are indicated at the top of each chromosome. Ten unmapped genes are shown on the right. The vertical green lines indicate the tandemly duplicated genes. The yellow bars with blue numbers on the chromosomes indicate the 11 predicted duplicated regions.

presented herein and the phylogeny described previously for *A. thaliana*, with all of the subgroups maintained in this multiple-species phylogeny. The only exceptions are the *AtR2R3-MYB* genes of subgroups S9, S10 and S25, which were each split into two branches (S9a, S9b, S10a, S10b, S25a and S25b) (Fig. 3-A, Appendix F).

Phylogenetic comparative analysis of R2R3-MYBs in different plant species revealed both considerable diversification and conservation of the MYB gene family in plants (He *et al.* 2016). In the present study, most of the clades (37 out of 46) were present both in *M. truncatula* and *A. thaliana*, indicating large scale conservation among the *M. truncatula* and *A. thaliana* MYB gene families. However, there was not equal representation of R2R3-MYB members among *M. truncatula* and *A. thaliana* within a given clade (Fig. 3-A, Appendix F). In some larger clades such as C1, C22, C24, C27, C30 and C42, two or more MtR2R3-MYBs were found to be putative orthologs of a single *A. thaliana* protein, which indicates that R2R3-MYBs in *M. truncatula* were duplicated after the divergence of *M. truncatula* and *A. thaliana*. Remarkably, some homologs were clustered by species within a clade (Fig. 3-A, Appendix F). For example, the *A. thaliana* ‘glucosinolate’ subgroup S12 did not contain any MtR2R3-MYBs, possibly because members involved in the regulation of glucosinolate biosynthesis are specific to the *Brassicaceae* family (Matus *et al.* 2008). In contrast, C25 did not include any *A. thaliana* R2R3-MYB proteins but only *M. truncatula* members. This suggests that the genes in C25 may have specialized roles that were either lost in *A. thaliana* or acquired in *M. truncatula* lineages after their divergence from the latest common ancestor (Dubos *et al.* 2010). Moreover, all of the members of C7, which have been identified as regulators of anthocyanin biosynthesis pathway, are from three monocot species, suggesting that C7 might be a novel monocot-specific clade that came into existence after the monocot/dicot divergence or that C7 was lost in dicots.

A. thaliana subgroups were categorized on the basis of conserved motifs present in the variable C-terminal region of the R2R3-MYB proteins. These conserved motifs are thought to play important roles in specifying functional divergence (Stracke *et al.* 2001; Soler *et al.* 2015). Using MEME, we searched for the conserved motifs in the C-terminal regions of the MtR2R3-MYB proteins belonging to each clade. One or more motifs outside of the MYB domain were shared by most members within the same clade, which provides further support for the clade definition based on phylogenetic analysis. For example, the 13 MtR2R3-MYBs clustered in C1 share motifs 2–5 in the C-terminus of the protein. In total, we detected 45 clade-specific motifs in the MtR2R3-MYB family (Table 1). Except for motif 1, which is adjacent to the R3 repeat and is conserved within 10 clades, most of the motifs were selectively present amongst specific

clades. This may indicate that motif 1 remained alongside the R2R3-MYB domain during evolution. The distribution of the conserved motifs within each clade is shown in Fig. 3-B.

3.5. Expression profile analysis of the R2R3-MYB genes in *M. truncatula* tissues

Microarray datasets from the *M. truncatula* reference genotype Jemalong A17 were downloaded from MtGEA and used to assess the transcript abundance profiles of MtR2R3-MYB-encoding genes in 10 major tissues: leaf, petiole, vegetative bud, stem, root, nodule, flower, pod, seed coat and seed. Because the probe sets corresponding to some genes were not represented in the dataset, we only examined the transcript abundance of 79 MtR2R3-MYB genes. In order to visualize the datasets, an expression heat map was generated. Approximately 20 of the MtR2R3-MYB-encoding genes exhibit low transcript abundance. In contrast, 7 genes (*MtMYB18*, *MtMYB54*, *MtMYB62*, *MtMYB70*, *MtMYB82*, *MtMYB84* and *MtMYB3R-1*) have a relatively high expression level in all 10 tissues (Fig. 4). In addition, some MtR2R3-MYB genes have preferential accumulation of transcripts in one or more tissues. For instance, *MtMYB110* and *MtMYB122* are exclusively expressed in flower, *MtMYB14*, *MtMYB18*, *MtMYB37*, *MtMYB65*, *MtMYB71*, *MtMYB72*, *MtMYB83*, *MtMYB85*, *MtMYB133* and *MtMYB149* showed the highest transcript accumulation in root, *MtMYB38* and *MtMYB60* in leaf and petiole, *MtMYB18* and *MtMYB54* in nodules, and *MtMYB20*, *MtMYB22*, *MtMYB52*, *MtMYB75*, *MtMYB103*, *MtMYB135*, *MtMYB136* and *MtMYB148* in the seed coat. The expression profiles of the duplicated paralogous genes were also examined based on the heat map and hierarchical clustering analysis. Duplicated genes with high sequence similarity have divergent expression profiles. These include one pair of tandem duplicated genes (*MtMYB13/MtMYB14*) and four pairs of segmental duplicated genes (*MtMYB21/MtMYB135*, *MtMYB29/MtMYB78*, *MtMYB112/MtMYB119* and *MtMYB113/MtMYB120*). Similar expression abundance profiles were observed for two paralogous gene pairs, *MtMYB20/MtMYB136* and *MtMYB38/MtMYB60*, which are highly expressed in the seed coat and leaf, respectively.

3.6. Expression profiles of R2R3-MYB genes in *M. truncatula* under abiotic and biotic stresses

Using the microarray datasets in MtGEA, we assessed the expression profiles of 27 candidate genes under two treatments: 0, 1, 2, 5, 10, and 24 h after 200 mmol L⁻¹ NaCl treatment; 0, 3, and 4 d after *Phytophthora* root rot infection. These genes were selected from the 8 functional clades (C11, C12, C16, C24, C28, C31, C34 and C39) that have members that are related to abiotic and biotic stress

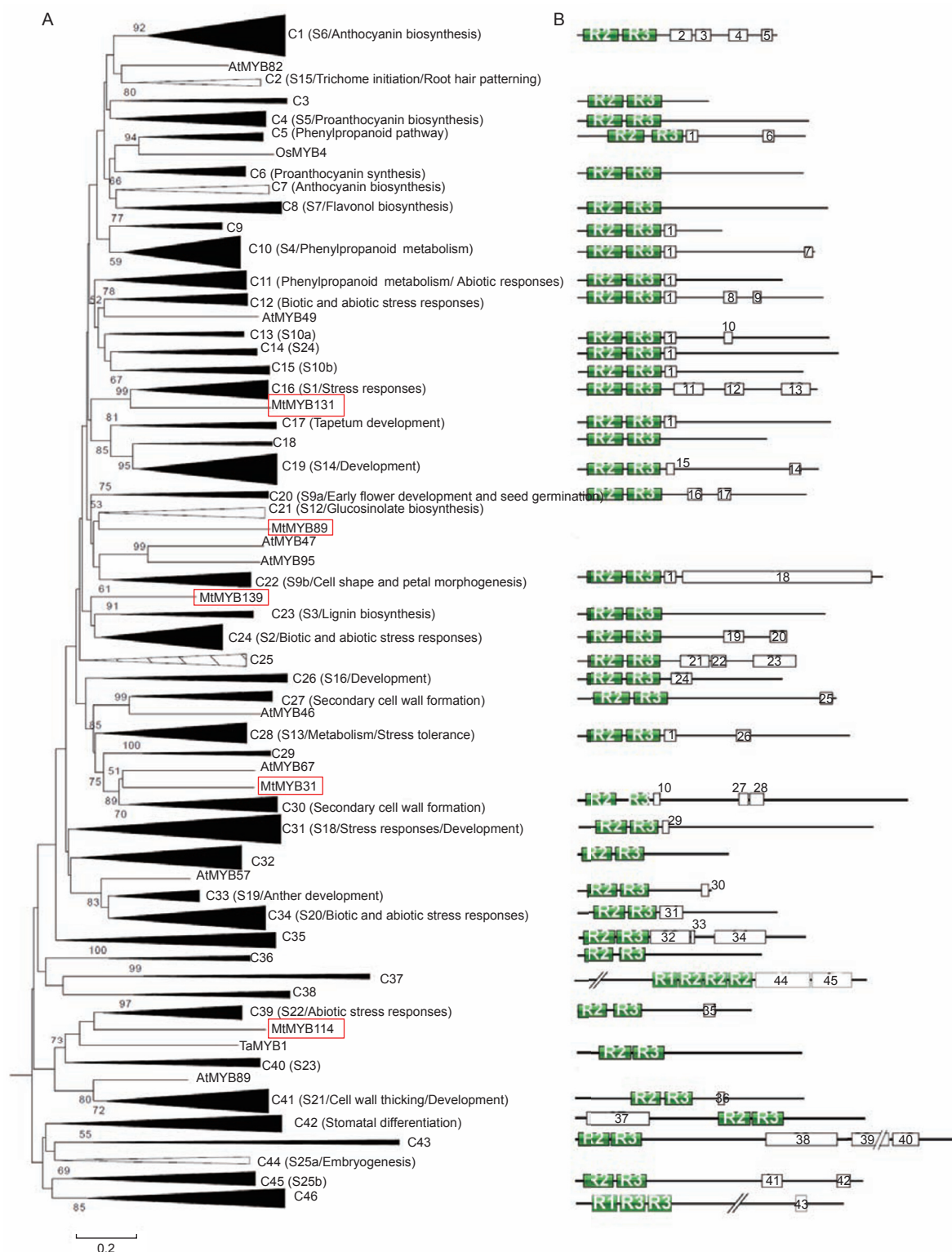


Fig. 3 Phylogenetic tree and subgroup designations of the R2R3-MYB proteins from *Medicago truncatula* (Mt), *Arabidopsis thaliana* (At) and other plant species. A, the unrooted phylogenetic tree was constructed based on the alignment of full length MYB sequences using MEGA v6.06 with 1000 bootstraps. Bootstrap support less than 50% were collapsed in the phylogenetic tree. R2R3-, 3R-, 4R-MYB proteins from *M. truncatula*, *A. thaliana* and other plant species are clustered into 46 subgroups (triangles), each designated with a subgroup number (e.g., C1). Five proteins did not fit well into any cluster (in red frames). Subgroups are represented as collapsed triangles, the black triangles indicate that the subgroup includes both MtMYBs and AtMYBs; the white and hatched triangles indicate that the subgroup excludes and includes MtMYBs, respectively. The clades are designated according to comparative analysis with *A. thaliana*. B, architecture of conserved protein motifs in 46 subgroups.

Table 1 MEME motif sequences in predicted MtR2R3-MYB proteins

Motif	E-value	Conserved sequences
1	2.00E-185	[IL][KQ]MGIDPVTHKP[LR]S[DE]X
2	3.50E-144	PKET[MI]KAHEVIKP[RQ]P[R]TFSTHS[PS]WL[NK][GK]KH
3	2.70E-16	N[FG][VS][TA][QPT][PIK][LV][LV][AIV]S[NST][EKN]DGNV[PS][RK][DE][RCS]D
4	1.80E-42	[SW][DE][SN]L[WL]NL[GD]E[QEH][VR][DNS][SN]EKIGS[CGS]S[SL]LQEE
5	4.50E-41	[VI][DS]D[SF]FWD[FS]N[LI]CD[FY][DN][SF]
6	3.40E-03	H[DH]HH[HQ]H
7	1.10E-06	[PQ]F[IF]DFL[GS][VT]G[ASV]
8	1.40E-13	[DM]IN[DN]HYN[AGP]NF[L]P[HD][QE]N[PQ][IM][PA]
9	4.20E-03	MSSSSSSSSTD
10	5.60E-09	[HD]DIVPSFT[PT]F[SY]
11	2.40E-90	[EG][EV][DE][KQ][EF]S[AR]SQ[LPR][IP][NP][KR]GQWERRLQTDI[HQ]MAK[KQ]AL[SN]EALS[LP]Q[KH]
12	7.50E-27	[QS][ST][FSL][CPST]YAS[SN][AT][DE]NI[AS]RL[KE][GK]WM[KR][NKS][PKS][IPE][KN][SE]
13	7.70E-20	S[PD][DS][AT]T[IL][LF]QDESKP[DE][VN][GI][SG]E[TI]MPFS[LM]LEKWLLD[ED][AG][SA]C[PQ]EKV
14	1.40E-12	LDY[SG][SL][DE]S[IK][EK]Q[LM][LS][NS]S
15	2.20E-06	K[QH]RK[ED]QQ[SALQ][QR][AR][RV]
16	3.00E-38	[TL][RS]HMAQWESARLEAEARL[SV]
17	3.00E-04	DYFL[KQR][IL]W[NH]S[ED]VG[DQ][AKS]FR
18	1.20E-105	YTKDSANLSHMAQWESARLEAEARLVRETNKLVQRQLEFESSKPQPPARLFLNKITPLIQPSLPPCLDILKAWQSS WSKPTTKDDSNNNNNKMHSMYAMMLSNDDLESPTSTLNFPGTLLPLPLSVPMTNNNIEPFNQSLP[FL]TNST NNDDDD[DY]DDD[NY]EIWKQFNLTKHNTGGHGHDDNNNTMLNFQDDDDIMAAVEAFRASGGFDNNINTPPIST NVLVQVLEGGDHQENLDGSMICNVNLEENKHYW
19	6.50E-11	DX[EV]D[SM][EW][EV]TM[AEP]X[F]I[DS]E[NSD]FW[TS]E[PT]Y[I]T[DIL]D[EN][TS][SY]
20	2.20E-13	[DC][DQ][GD]M[DG]FW[YHV]D[VIL][FLY][IA][RK][SAT][EG][ED][SIQ][TI]EL[PL]E[FL]
21	2.00E-09	STQ[GD]R[DC][IV]S[NF][YL]Q[IV]T[PL]P[AD][SN]SQ[I]T[SL][DN][INT][NT]GP[LF]SP[FL]SSSSE[FIL]SST
22	1.10E-06	DDFGFLD[SG]FI[DEK][DY]V[DG][EI][SK][FS]W[LD][HE]
23	5.70E-50	[VN][MT][DN][NR]DF[GE]SFLDAY[NK]EST[VI]DSFWTQPYE[AV]DM[SY]HV[PS][RST][QE][LM]L[AIT][PS]L[PS] ME[SP]E[YN]FSIVYD[DK]DIW
24	6.30E-12	H[KT]Q[F]I]QYASSSDTMNS[LS][LP]S[LT]QK[LY]A[IT]QDP
25	3.10E-23	G[ED]W[DN][LF]E[DEG]LM[KQ]D[VI]S[SY]F[PL][FP]
26	1.90E-24	[QP][FC][ED][QL][NS]QFG[LNT]NSSY[F]G[F]S[MD][PN]
27	1.10E-35	[IM]N[PQ][IS][ES]LMQQYDQYHD
28	2.00E-44	LVE[LH][ED]A[TI]VP[KE]L[VL]N[DH]QSLE[DN]YAC
29	2.80E-06	R[AQ]GLP[L]Y[PS][PD]E[HIV]
30	6.60E-11	[FL]W[NT][DAV]EN[M]I[LCW][FL][LF]Q[QH]
31	1.20E-43	[KR]QLKCDVNS[KQ]QF[KR]D[TAV][LM]R[YH][LV]W[M]PRL[VIL]E[RQ]IQ[APS][AS][AGP][AQS][TAF][APST][DAT]T
32	3.80E-230	KLGLSPFPDNMEPDHKLNSQQVEDSQEDEFNIPQLKFRKYPSIFDEINEKLLDVPNMFYNN
33	2.10E-11	VGSTST
34	7.20E-157	IDR[VG]SSIIGTSHIHPIALPSPCSYLPSPMMLDE[KN]GKKPISLEDD[VF]EANLDDFMDLIDFSNLEDDVEAGS RLFDNDDI
35	5.50E-06	[EK][FL][LMRW][SAD][VM][ML][QR][EGR][MV][AKR][ENRT]EVR[SEN]Y[MV]
36	1.10E-06	[PQ]F[IF]DFL[GS][VT]G[ASV]
37	8.90E-129	[DAG][AIT][HWY][NTD][NFS][GS][SQ]Q[RNS][EL]F[ILQ][SY][AST][REQ][NT][PS][MH][ISV][FQ][TASV][PST][NP] [NHSY][NL][KS][IMT][ES][AVD][MIV][HY][GE][CRP][LV][NH][TS][SEGR][KE]G[IT][WR][DE][LF]S[KTQ][KNQ][N]IL F[QRP]Y[GD]ETSQ[PSY][RMQW]VSP[SD]LS[PT]S[LV][VA][YS]D[AEG][HN][PQR]S[VL][SL][VI][KN][PS][EKR][LH] Q[GAI][DGN][LIT][SF][FCRS][NYS][GV]G[FIL][GRV]N[EKLN][PS][QR][EH]N[DN]
38	4.30E-43	MLPPPQISDQELDEIAKLGYASDLVGSEELTEGSSATRALLPNYPQTPNQAKTPLRTPQRTPASKGDAIMME AENLARLRESQTPLLGGENPELHPSDFSGVTPKKKEIHTP
39	3.20E-117	MTPARDGFSFGMTPKGTPLRDELHINEDMEMRDSAKLELRQADMKKSLRSGLSSLPQPKNEYQIVMQPIQED VEEPKEIEEDMSDRMAREKAEERQALLRKRSKVLQRELPRPPASLELIRNSLIRADGDKSSFVPPTPI EQADEMIRKELLALLEHDNAKYPLEEIANKERKKGSKRAANGPAIPVIEDYQEDEMKSADQLIKDEAQYLREAMGH ENDSLEEFVEAHTTCINDLMYFDTRNAY
40	8.20E-10	VAGNAEKLAALQNEFENVRSKLDDGKEKLIRLEKKVTTFTQ
41	3.90E-07	[YP][PS][GKR][GS]TLLQ[EK]YIKSLNLD[EKQ]NPP[KR][DE][YF][RT][RK][KR][SP]S[AT][KN]A
42	7.30E-25	GMEF[FM][PL][EG][VI][PE]VK[QK]E[IL]DLME[MK]IY
43	1.00E-12	[PG][LCS]V[AR][AD][KA][ED][THP][IET][KN][RIL][SN][VA][SE][KEV][TN][ID][LDV][VD][SL][PS]
44	2.40E-33	EAKKKORSQSLSRNFVDRESERPTLAVNDLIPLQMVVPRSDVGAENLRKKRKRKSSHQKERSKKHAKHAAGLCIEEVQ DADPKKERPKRHAKKKVPFCPEEQDIVPKKERPKRH
45	1.70E-97	ARICPEEGQFIAAYSDDMKMKSARGVPLFEQSNVHNMRSKRHAKKAQIHLKEVENIAFNDKVKTCSKISESQDGD DITLACWRNKLKKLSNCTKNASQTSFSMPKIVSKQVGNQIPSNQDGLSLTCGTGGTEDLLIQTEVDSSEQVR KPESENAARKPEDAHNMNGDGNKPLQVYVRRKTKKWSSEDTGCRACSPSKLKKGSMLLHGNKPAIISDGSEP SMKVVVEETVLHGGVAEAETNINVV

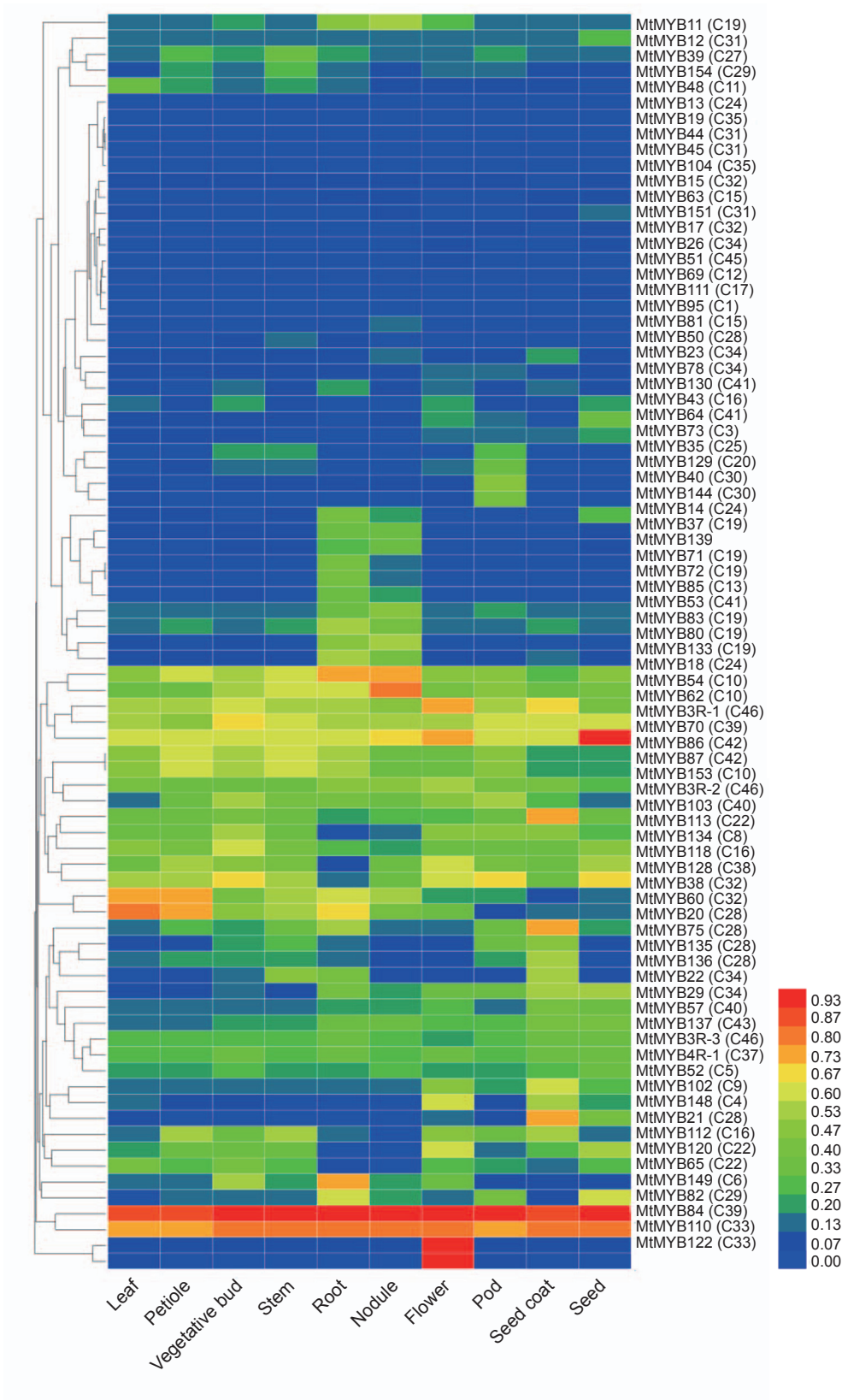


Fig. 4 Heat map representation of the expression profiles of the *MtMYBs* among different tissues. Heat map showing the hierarchical clustering of 79 *MtR2R3-MYB* genes based on expression profiles in ten tissues (leaf, petiole, vegetative bud, stem, root, nodule, flower, pod, seed coat and seed). Higher and lower levels of transcript accumulation are indicated by red and blue, respectively, and the median level is indicated by green. Microarray data were obtained from the *Medicago truncatula* Gene Expression Atlas (MtGEA) database, and the heat map was generated using Heml (ver. 1.0.3.3).

response (Fig. 3-A). In summary, most of the selected *MtR2R3-MYBs* have variable expression profiles during the course of stress treatments (Fig. 5). Four genes from C34 (*MtMYB22*, *MtMYB23*, *MtMYB29* and *MtMYB78*), and *MtMYB14* from C24 were up-regulated by both NaCl and *Phymatotrichum*, whereas *MtMYB48* (C11) and *MtMYB13* (C24) were induced by NaCl treatment and *Phymatotrichum* root rot infection, respectively. In contrast, *MtMYB20*, *MtMYB75* and *MtMYB136* (C28) were down-regulated by both treatments.

To validate the microarray data, 5 genes (*MtMYB14*, *MtMYB22*, *MtMYB23*, *MtMYB29*, and *MtMYB48*) were selected for qRT-PCR analysis of gene expression under

various stress treatments including salinity, drought, cold, and ABA. Selection of these genes was based on their statistically significant up-regulated expression in the microarray analysis. As shown in Fig. 6, the expression patterns of *R2R3-MYB* genes under salt treatment in the qRT-PCR and microarray analyses generally follow the same trends, but the qRT-PCR method is more sensitive than microarray analysis for detecting transcript expression. The results of qRT-PCR showed that *MtMYB23*, *MtMYB29* and *MtMYB48* are up-regulated in response to salinity, drought, cold, and ABA stresses, whereas *MtMYB14* and *MtMYB22* are both up-regulated in response to salinity and ABA, but down-regulated in response to drought and cold (Appendix G).

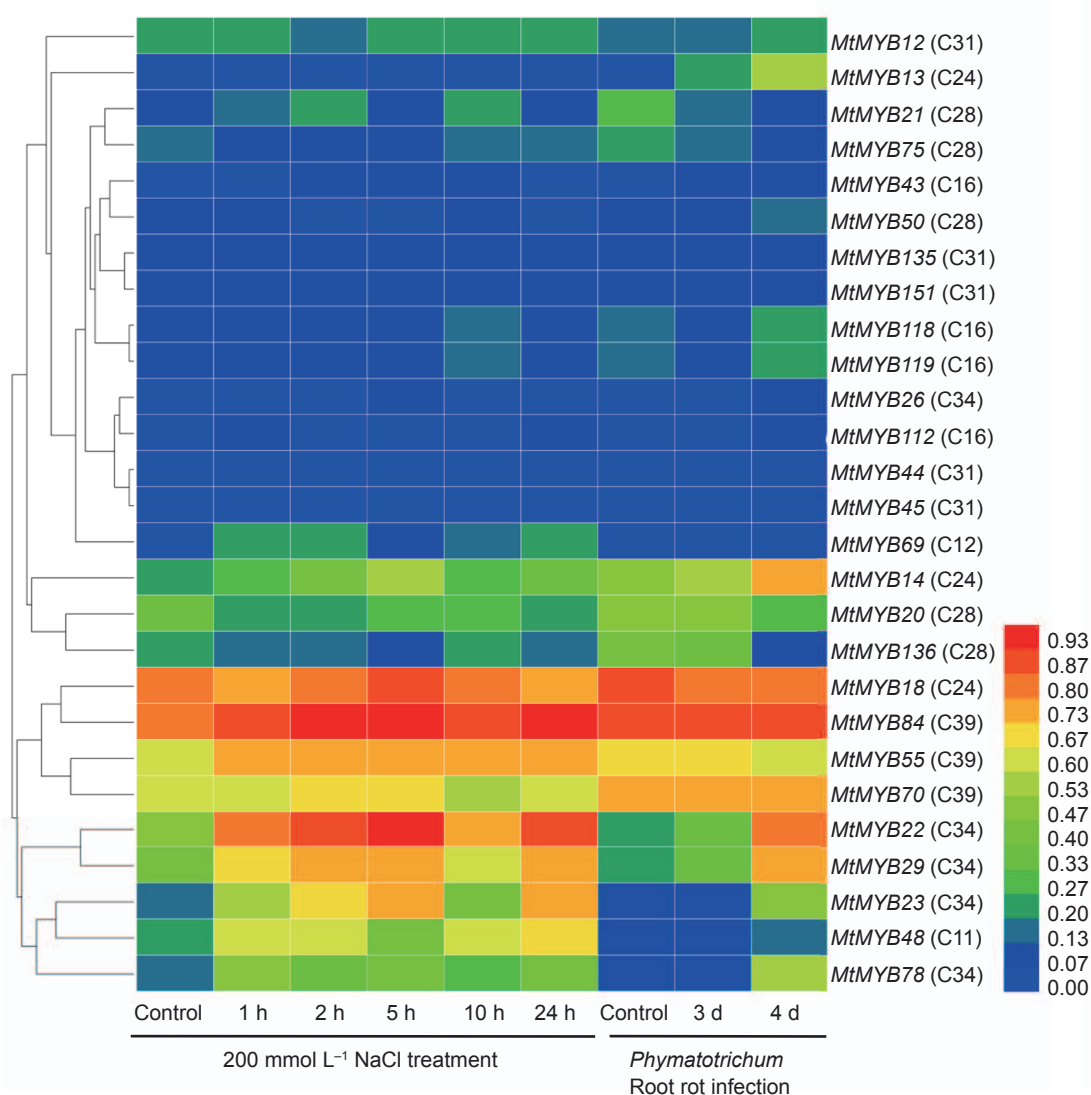


Fig. 5 Expression of *MtR2R3-MYB* genes in response to salt treatment and fungal infection. Heat map showing the changes in expression level of 27 *MtR2R3-MYB* genes at different time points after treatment with 200 mmol L⁻¹ NaCl (0, 1, 2, 5, 10, 24 h) and *Phymatotrichum* (0, 3, 4 day) treatment in root. Higher and lower levels of transcript accumulation are indicated by colored red and blue, respectively, and the median level is indicated by green. Microarray data were obtained from The *Medicago truncatula* Gene Expression Atlas (MtGEA) database, and the heat map was generated using Heml (ver. 1.0.3.3).

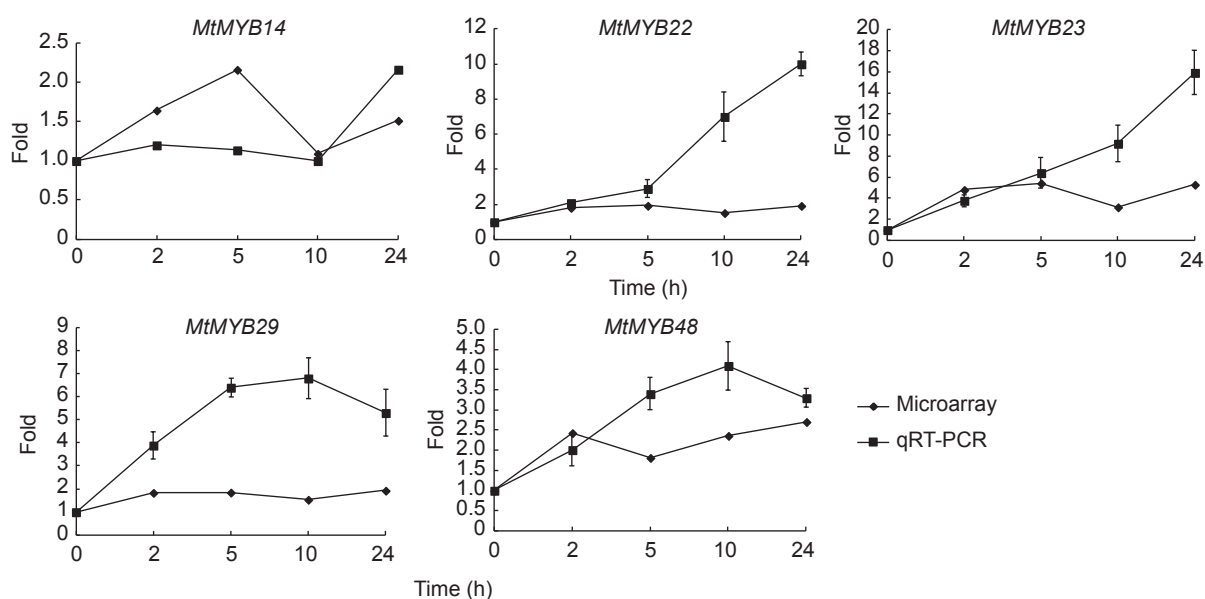


Fig. 6 qRT-PCR validation of microarray results. Fold changes in the expression of five *R2R3-MYB* genes in roots under salt treatment were obtained from qRT-PCR experiments and microarray datasets. Data represent fold change in expression level at each time-point relative to 0 h. Error bars indicate SE.

4. Discussion

4.1. Characterization of the *MtR2R3-MYB* family genes

The *R2R3-MYB* TF family has been extensively studied in a number of species, and there are at least 125 and 244 members in *A. thaliana* and soybean, respectively. However, a comprehensive investigation of the *R2R3-MYB* gene family in *M. truncatula* is still lacking. In the present study, a total of 155 *MtR2R3-MYB*s were identified throughout the genome of *M. truncatula*, which is 0.64 times that of soybean. The fact that the soybean genome is almost two times larger (~970 Mb in soybean and ~412 Mb in *M. truncatula*) with a similar gene density (one gene per 5.8–7.2 kb) as *M. truncatula* (Shoemaker et al. 2006), might at least partially explain the higher number of soybean *R2R3-MYB* genes (Schmutz et al. 2010; Qin et al. 2016).

TFs within a gene family generally harbor some conserved domains that are important for conserved regulatory functions. The alignment of 155 typical *MtR2R3-MYB* DNA-binding domains (Appendix D) revealed that both R2 and R3 consist of three α -helices and that there is a remarkable excess of amino acid substitutions in the first two helices of the *MtR2R3-MYB* DNA-binding domain compared with helix 3. This is consistent with previous findings in rice and maize (Jia et al. 2004; Du et al. 2012a), and indicates that the helix 3 was more highly conserved than the first 2 helices during evolution. Previous studies have demonstrated

that α -helix 3 in each repeat is involved in direct interaction with the major groove of target DNA, and mutations occurring in helix 3 often result in the loss of function (Ogata et al. 1994; Kelemen et al. 2015). For instance, the presence of conserved [DE]Lx₂[RK]x₃Lx₆x₃R residues in the R3 repeat, which are critical for interaction with bHLH TFs, is typical for anthocyanin- and PA-regulating MYB TFs (Hichri et al. 2011; Huang et al. 2013).

Despite the divergence of the C-terminal regions outside the MYB domain, there are conserved motifs among *R2R3-MYB* proteins within specific clades (Fig. 3-B), further supporting the clade definition based on phylogenetic analysis. Although the functions of most of the motifs have not been elucidated, it is likely that some may have a role in the transcriptional regulation of target genes (Li et al. 2006). For instance, *AtMYB*s and other landmark proteins in clade C1 possess the conserved motif 2 at the C-terminus (Fig. 3-B, Table 1) and are known to be involved in anthocyanin biosynthesis (Peel et al. 2009; Cavallini 2014). Thus, the *MtR2R3-MYB*s in this clade might play roles in the anthocyanin biosynthetic pathway. Remarkably, members belonging to subgroup 4 of *A. thaliana* were reported to harbor a conserved repressive motif and a putative zinc-finger domain in the C-terminus (Jin and Martin 1999; Du et al. 2009). Many other *R2R3-MYB* transcriptional repressors, such as *GmMYBZ2* (Zhou et al. 2011), *EgMYB1* (Legay et al. 2010), *TaMYB4* (Ma et al. 2011) and *OsMYB2* (Yang et al. 2012), also contain this subgroup 4-specific motif. In the present study, we identified three orthologous *MtR2R3-*

MYBs (MtMYB153, MtMYB54 and MtMYB62) that contain the same conserved domain in the C-terminal region, which may suggest that these MtR2R3-MYBs also function as repressors in *M. truncatula*.

4.2. Phylogenetic analysis and putative functions of MtR2R3-MYB genes

Previous comparative phylogenetic analyses of R2R3-MYBs in model plant species revealed both considerable diversification and conservation of the *R2R3-MYB* gene family in plants (Wilkins et al. 2009; Du et al. 2012a, b). In general, orthologs within a clade share similar gene structures and functions, indicating recent common evolutionary origins. However, species-specific clades also existed which were gained or lost during expansion of the *MYB* gene family, resulting in functional divergence. Although the roles of most *MtR2R3-MYBs* remain to be illuminated, members within each clade are assumed to have recent common evolutionary origins and also conserved functions.

According to our analysis, in most clades MtR2R3-MYBs are clustered with orthologs from other plant species (Fig. 3, Appendix F), suggesting the functional conservation of R2R3-MYBs between species. For example, 12 MtR2R3-MYBs belong to clade 1, and among these, there are two *Arabidopsis* proteins that regulate anthocyanin biosynthesis, AtMYB75/PAP1 and AtMYB90/PAP2. This implies that the functions of *MtR2R3-MYB* genes in this clade are related to anthocyanin biosynthesis. This speculation is supported by the known functions of clade 1 members MtMYBA (MtMYB101 in this study) (Carletti et al. 2013) and MtLAP1 (MtMYB152) (Peel et al. 2009), both of which have a role in leaf anthocyanin accumulation. Plants adapt to unfavorable environmental conditions by dynamic reprogramming of their stress-responsive genes (Walther et al. 2007). As one of the largest TF groups in plants, R2R3-MYB proteins play an important role in these responses. Seven clades, 12, 16, 24, 28, 31, 34 and 39, contain many notable R2R3-MYB proteins that are known to have important roles in these processes. For example, AtMYB44 in subgroup 22 regulates ABA-mediated stomatal movement to confer tolerance to abiotic stresses (Jung et al. 2008) and three other AtR2R3-MYBs in this subgroup (AtMYB70, AtMYB73 and AtMYB77/AtMYB72) are likely associated with stress responses (Kim et al. 2013; Baldoni et al. 2015). Thirty-eight R2R3-MYBs belong to clades that have genes with known roles in stress response, and thus may also play roles in the response or tolerance to stress conditions.

4.3. Gene duplication and evolution of MtMYB genes

Gene duplication is believed to play an important role in

gene family expansion in plants (Nuruzzaman et al. 2010), either in the form of segmental duplication (blocks of genes on chromosomes), tandem duplication (an individual gene), or transposition events (He et al. 2016). In *A. thaliana* (as well as in soybean and poplar), both ancient and recent genome duplications have led to large-scale expansion of the *R2R3-MYB* gene family (Kohler et al. 2008). In the present study, we observed that both segmental duplication and tandem duplication events contributed to the *R2R3-MYB* gene family expansion in *M. truncatula*. Similar expansion events were also found in other plant species, for instance in maize and *Gossypium raimondii* (Du et al. 2012a; He et al. 2016). However, compared with the large-scale syntenic blocks containing *R2R3-MYB* genes in soybean, which are derived from a recent polyploidization event (Du et al. 2012b), the segmental duplication of *MtR2R3-MYB* genes seems to be considerably limited. Recent research suggests that the lineage leading to *Glycine* experienced another polyploidy and more recent genome duplication events occurred up to 13 million years ago (Schmutz et al. 2010; Lin et al. 2014). Shin et al. (2008) found that there was no polyploidization event after the *Medicago*-soybean split, but a series of tandem duplications contributed to the expansion of lipoxygenase family in *M. truncatula*. In addition, the presence of three tandem arrays of *MtMYB71/MtMYB72*, *MtMYB118/MtMYB119* and *MtMYB137/MtMYB138* (located on the recently duplicated blocks 9, 10 and 1, respectively) suggest that a tandem duplication of the ancestor of these genes occurred prior to the most recent segmental duplication.

Over the course of evolution, many duplicate genes (paralogs) are quickly erased, losing functionality and finally being removed by deletion (Panchy et al. 2016). Microarray data show that some *R2R3-MYB* paralogs (*MtMYB38/MtMYB60*, *MtMYB86/MtMYB87*, *MtMYB20/MtMYB136*) exhibit similar expression profiles, and thus show potential functional redundancy during tissue development (Fig. 4). Divergence of expression patterns of other duplicated paralogous *MYB* genes may reflect the preservation of duplicated genes through sub- and neofunctionalization (Lin et al. 2014). For instance, *MtMYB78* is expressed at low levels in the seed coat and seed tissues, whereas its counterpart, *MtMYB29*, is preferentially expressed in these tissues.

The degree of genetic redundancy within *R2R3-MYB* subgroups can also be assessed by the response of those genes to environmental stresses. Two duplicated genes within the subgroup C34, *MtMYB29* and *MtMYB78*, have similar expression profiles in response to NaCl and *Phytophthora* treatment (Fig. 5), indicating functional redundancy. Based on qRT-PCR analyses of gene expression, the closely related *MtMYB22* and *MtMYB23* genes are both up-regulated significantly in response to NaCl and ABA but down-regulated by mannitol. However, this paralogous gene

pair behaved oppositely when subjected to 4°C treatment, indicating that these genes have functionally diverged after duplication. The soybean ortholog of this pair, *GmMYB76*, is induced by NaCl treatment but has no significant response to 4°C and ABA treatments (Liao *et al.* 2008). The *A. thaliana* homolog, *AtMYB2*, is induced by drought and ABA treatments (Abe *et al.* 2003). These results indicate that the functions of homologous *R2R3-MYBs* appear highly, but not absolutely, conserved across plant species.

In plants, there was considerable expansion of the *R2R3-MYB* family, but expansion of *3R-MYB* genes was relatively limited. We detected only five *3R-MYB* genes orthologous to *A. thaliana* in the *M. truncatula* genome, and the same number was also identified in other plant lineages, such as *A. thaliana* (Stracke *et al.* 2001), *V. vinifera* (Matus *et al.* 2008) and *P. trichocarpa* (Wilkins *et al.* 2009). The *3R-MYBs* typify the vertebrate *MYB* family (Jin and Martin 1999), and they have also been retained in many plant lineages during gene expansion (Braun and Grotewold 1999), suggesting that the *3R-MYB* genes may represent an ancient and evolutionarily conserved gene family. Moreover, *3R-MYBs* have been shown to control the cell cycle, which is evolutionarily conserved in animals and plants (Ito *et al.* 2001; Haga *et al.* 2007). The fact that the *R2R3-MYB* family in plant is larger and more diverse suggests that they may fulfill a wide range of diverse functions within this kingdom.

5. Conclusion

We identified 155 *MtR2R3-MYB* genes from the *M. truncatula* genome and comprehensively analyzed their domain architecture, chromosome distribution, phylogeny, and gene expression patterns. Additionally, phylogenetic relationships and gene expression profiles were used to predict the potential functions of these *MtR2R3-MYB* genes. The results from this study provided a basic understanding of the characterization of *MtR2R3-MYB* genes and will facilitate future research aimed at elucidating the functions of *R2R3-MYB* genes in *M. truncatula*.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (31372362). We thank Dr. Wang Hai from Biotechnology Research Institute, Chinese Academy of Agricultural Sciences, for critical reading of the manuscript.

Appendix associated with this paper can be available on <http://www.ChinaAgriSci.com/V2/En/appendix.htm>

References

- Abe H, Urao T, Ito T, Seki M, Shinozaki K, Yamaguchi-Shinozaki K. 2003. *Arabidopsis* AtMYC2 (bHLH) and AtMYB2 (MYB) function as transcriptional activators in abscisic acid signaling. *The Plant Cell*, **15**, 63–78.
- Albert N W. 2015. Subspecialization of *R2R3-MYB* repressors for anthocyanin and proanthocyanidin regulation in forage legumes. *Frontiers in Plant Science*, **6**, 13.
- Ambawat S, Sharma P, Yadav N R, Yadav R C. 2013. MYB transcription factor genes as regulators for plant responses: An overview. *Physiology and Molecular Biology of Plants*, **19**, 307–321.
- Bailey T L, Elkan C. 1994. Fitting a mixture model by expectation maximization to discover motifs in bipolymers. *International Conference on Intelligent Systems for Molecular Biology*, **2**, 28–36.
- Baldoni E, Genga A, Cominelli E. 2015. Plant MYB transcription factors: Their role in drought response mechanisms. *International Journal of Molecular Sciences*, **16**, 15811–15851.
- Baumann K, Perez-Rodriguez M, Bradley D, Venail J, Bailey P, Jin H L, Koes R, Roberts K, Martin C. 2007. Control of cell and petal morphogenesis by *R2R3 MYB* transcription factors. *Development*, **134**, 1691–1701.
- Berardini T Z, Reiser L, Li D H, Mezheritsky Y, Muller R, Strait E, Huala E. 2015. The *Arabidopsis* Information Resource: Making and mining the “gold standard” annotated reference plant genome. *Genesis*, **53**, 474–485.
- Bonthala V S, Mayes K, Moreton J, Blythe M, Wright V, May S T, Massawe F, Mayes S, Twycross J. 2016. Identification of gene modules associated with low temperatures response in Bambara groundnut by network-based analysis. *PLOS ONE*, **11**, doi: 10.1371/journal.pone.0148771
- Braun E L, Grotewold E. 1999. Newly discovered plant *c-myb*-like genes rewrite the evolution of the plant *myb* gene family. *Plant Physiology*, **121**, 21–24.
- Carletti G, Lucini L, Busconi M, Marocco A, Bernardi J. 2013. Insight into the role of anthocyanin biosynthesis-related genes in *Medicago truncatula* mutants impaired in pigmentation in leaves. *Plant Physiology and Biochemistry*, **70**, 123–132.
- Cavallini E. 2014. Functional diversification of grapevine MYB5a and MYB5b in the control of flavonoid biosynthesis in a petunia anthocyanin regulatory mutant. *Plant & Cell Physiology*, **55**, 517–534.
- Chai G, Wang Z, Tang X, Yu L, Qi G, Wang D, Yan X, Kong Y, Zhou G. 2014. *R2R3-MYB* gene pairs in *Populus*: Evolution and contribution to secondary wall formation and flowering time. *Journal of Experimental Botany*, **65**, 4255–4269.
- Deng W, Wang Y, Liu Z, Cheng H, Xue Y. 2014. Heml: A toolkit for illustrating heatmaps. *PLOS ONE*, **9**, doi: 10.1371/journal.pone.0111988
- Du H, Feng B R, Yang S S, Huang Y B, Tang Y X. 2012a. The *R2R3-MYB* transcription factor gene family in maize. *PLoS*

- ONE, **7**, doi: 10.1371/journal.pone.0037463
- Du H, Liang Z, Zhao S, Nan M G, Tran L S P, Lu K, Huang Y B, Li J N. 2015. The evolutionary history of R2R3-MYB proteins across 50 eukaryotes: New insights into subfamily classification and expansion. *Scientific Reports*, **5**, 16.
- Du H, Yang S S, Liang Z, Feng B R, Liu L, Huang Y B, Tang Y X. 2012b. Genome-wide analysis of the MYB transcription factor superfamily in soybean. *BMC Plant Biology*, **12**, 1–22.
- Du H, Zhang L, Liu L, Tang X F, Yang W J, Wu Y M, Huang Y B, Tang Y X. 2009. Biochemical and molecular characterization of plant MYB transcription factor family. *Biochemistry*, **74**, 5–16.
- Dubos C, Stracke R, Grotewold E, Weisshaar B, Martin C, Lepiniec L. 2010. MYB transcription factors in *Arabidopsis*. *Trends in Plant Science*, **15**, 573–581.
- Feng S, Xu Y, Yang L, Sun S, Wang D, Chen X. 2015. Genome-wide identification and characterization of R2R3-MYB transcription factors in pear. *Scientia Horticulturae*, **197**, 176–182.
- Finn R D, Coghill P, Eberhardt R Y, Eddy S R, Mistry J, Mitchell A L, Potter S C, Punta M, Qureshi M, Sangrador-Vegas A, Salazar G A, Tate J, Bateman A. 2016. The Pfam protein families database: Towards a more sustainable future. *Nucleic Acids Research*, **44**, D279–D285.
- Franco-Zorrilla J M, Lopez-Vidriero I, Carrasco J L, Godoy M, Vera P, Solano R. 2014. DNA-binding specificities of plant transcription factors and their potential to define target genes. *Proceedings of the National Academy of Sciences of the United States of America*, **111**, 2367–2372.
- Geethalakshmi S, Barathkumar S, Prabu G. 2015. The MYB transcription factor family genes in sugarcane (*Saccharum* sp.). *Plant Molecular Biology Reporter*, **33**, 512–531.
- Gocal G F W, Poole A T, Gubler F, Watts R J, Blundell C, King R W. 1999. Long-day up-regulation of a *GAMYB* gene during *Lolium temulentum* inflorescence formation. *Plant Physiology*, **119**, 1271–1278.
- He Q L, Jones D C, Li W, Xie F L, Ma J, Sun R R, Wang Q L, Zhu S J, Zhang B H. 2016. Genome-wide identification of R2R3-MYB genes and expression analyses during abiotic stress in *Gossypium raimondii*. *Scientific Reports*, **6**, 22980.
- Hall B G. 2013. Building phylogenetic trees from molecular data with MEGA. *Molecular Biology and Evolution*, **30**, 1229–1235.
- Haga N, Kato K, Murase M, Araki S, Kubo M, Demura T, Suzuki K, Muller I, Voss U, Jurgens G, Ito M. 2007. R1R2R3-MYB proteins positively regulate cytokinesis through activation of KNOLLE transcription in *Arabidopsis thaliana*. *Development*, **134**, 1101–1110.
- Hichri I, Barrieu F, Bogs J, Kappel C, Delrot S, Lauvergeat V. 2011. Recent advances in the transcriptional regulation of the flavonoid biosynthetic pathway. *Journal of Experimental Botany*, **62**, 2465–2483.
- Hou X J, Li S B, Liu S R, Hu C G, Zhang J Z. 2014. Genome-wide classification and evolutionary and expression analyses of citrus MYB transcription factor families in sweet orange. *PLOS ONE*, **9**, doi: 10.1371/journal.pone.0112375
- Huang W J, Sun W, Lv H Y, Luo M, Zeng S H, Pattanaik S, Yuan L, Wang Y. 2013. A R2R3-MYB transcription factor from *Epimedium sagittatum* regulates the flavonoid biosynthetic pathway. *PLOS ONE*, **8**, doi: 10.1371/journal.pone.0070778
- Ito M, Araki S, Matsunaga S, Itoh T, Nishihama R, Machida Y, Doonan J H, Watanabe A. 2001. G2/M-phase-special transcription during the plant cell cycle is mediated by c-Myb-like transcription factors. *The Plant Cell*, **13**, 1891–1905.
- Jia L, Clegg M T, Jiang T. 2004. Evolutionary dynamics of the DNA-binding domains in putative R2R3-MYB genes identified from rice subspecies *indica* and *japonica* genomes. *Plant Physiology*, **134**, 575–585.
- Jiang C H, Gu J Y, Chopra S, Gu X, Peterson T. 2004. Ordered origin of the typical two- and three-repeat Myb genes. *Gene*, **326**, 13–22.
- Jin H L, Martin C. 1999. Multifunctionality and diversity within the plant MYB-gene family. *Plant Molecular Biology*, **41**, 577–585.
- Jung C, Seo J S, Han S W, Koo Y J, Kim C H, Song S I, Nahm B H, Choi Y D, Cheong J J. 2008. Overexpression of *AtMYB44* enhances stomatal closure to confer abiotic stress tolerance in transgenic *Arabidopsis*. *Plant Physiology*, **146**, 623–635.
- Katiyar A, Smita S, Lenka S K, Rajwanshi R, Chinnusamy V, Bansal K C. 2012. Genome-wide classification and expression analysis of MYB transcription factor families in rice and *Arabidopsis*. *BMC Genomics*, **13**, 1–19.
- Kelemen Z, Sebastian A, Xu W J, Grain D, Salsac F, Avon A, Berger N, Tran J, Dubreucq B, Lurin C, Lepiniec L, Contreras-Moreira B, Dubos C. 2015. Analysis of the DNA-binding activities of the *Arabidopsis* R2R3-MYB transcription factor family by one-hybrid experiments in yeast. *PLOS ONE*, **10**, doi: 10.1371/journal.pone.0141044
- Kim J H, Nguyen N H, Jeong C Y, Nguyen N T, Hong S W, Lee H. 2013. Loss of the R2R3 MYB, *AtMyb73*, causes hyper-induction of the *SOS1* and *SOS3* genes in response to high salinity in *Arabidopsis*. *Journal of Plant Physiology*, **170**, 1461–1465.
- Kim J, Yang J Y, Yang R H, Sicher R C, Chang C, Tucker M L. 2016. Transcriptome analysis of soybean leaf abscission identifies transcriptional regulators of organ polarity and cell fate. *Frontiers in Plant Science*, **7**, 16.
- Kohler A, Rinaldi C, Duplessis S, Baucher M, Geelen D, Duchaussoy F, Meyers B C, Boerjan W, Martin F. 2008. Genome-wide identification of NBS resistance genes in *Populus trichocarpa*. *Plant Molecular Biology*, **66**, 619–636.
- Kranz H D, Denekamp M, Greco R, Jin H, Leyva A, Meissner R C, Petroni K, Urzainqui A, Bevan M, Martin C, Smeekens S, Tonelli C, Paz-Ares J, Weisshaar B. 1998. Towards functional characterisation of the members of the R2R3-MYB gene family from *Arabidopsis thaliana*. *The Plant Journal*, **16**, 263–276.
- Lau S, Schwarzsacher T, Rofina Yasmin O, Harikrishna J A. 2015. dsRNA silencing of an R2R3-MYB transcription factor affects flower cell shape in a *Dendrobium* hybrid. *BMC Plant Biology*, **15**, 1–14.

- Legay S, Sivadon P, Blervacq A S, Pavy N, Baghdady A, Tremblay L, Levasseur C, Ladouce N, Lapierre C, Seguin A, Hawkins S, Mackay J, Grima-Pettenati J. 2010. *EgMYB1*, an R2R3 MYB transcription factor from eucalyptus negatively regulates secondary cell wall formation in *Arabidopsis* and poplar. *New Phytologist*, **188**, 774–786.
- Li C, Ng C K Y, Fan L M. 2015. MYB transcription factors, active players in abiotic stress signaling. *Environmental and Experimental Botany*, **114**, 80–91.
- Li J, Yang X, Wang Y, Li X, Gao Z, Pei M, Chen Z, Qu L J, Gu H. 2006. Two groups of MYB transcription factors share a motif which enhances trans-activation activity. *Biochemical and Biophysical Research Communications*, **341**, 1155–1163.
- Li X, Xue C, Li J, Qiao X, Li L, Yu L a, Huang Y, Wu J. 2016. Genome-wide identification, evolution and functional divergence of MYB transcription factors in Chinese white pear (*Pyrus bretschneideri*). *Plant and Cell Physiology*, **57**, 824–847.
- Liao Y, Zou H F, Wang H W, Zhang W K, Ma B, Zhang J S, Chen S Y. 2008. Soybean *GmMYB76*, *GmMYB92* and *GmMYB177* genes confer stress tolerance in transgenic *Arabidopsis* plants. *Cell Research*, **18**, 1047–1060.
- Lin Y X, Cheng Y, Jin J, Jin X L, Jiang H Y, Yan H W, Cheng B J. 2014. Genome duplication and gene loss affect the evolution of heat shock transcription factor genes in Legumes. *PLOS ONE*, **9**, doi: 10.1371/journal.pone.0102825
- Lipsick J S. 1996. One billion years of Myb. *Oncogene*, **13**, 223–235.
- Liu J, Osbourn A, Ma P. 2015. MYB transcription factors as regulators of phenylpropanoid metabolism in plants. *Molecular Plant*, **8**, 689–708.
- Liu Z, Luan Y S, Li J B, Yin Y L. 2016. Expression of a tomato MYB gene in transgenic tobacco increases resistance to *Fusarium oxysporum* and *Botrytis cinerea*. *European Journal of Plant Pathology*, **144**, 607–617.
- Livak K J, Schmittgen T D. 2001. Analysis of relative gene expression data using real-time quantitative PCR and the $2^{-\Delta\Delta C_T}$ method. *Methods*, **25**, 402–408.
- Ma Q H, Wang C, Zhu H H. 2011. TaMYB4 cloned from wheat regulates lignin biosynthesis through negatively controlling the transcripts of both cinnamyl alcohol dehydrogenase and cinnamoyl-CoA reductase genes. *Biochimie*, **93**, 1179–1186.
- Martin C, Paz-Ares J. 1997. MYB transcription factors in plants. *Trends in Genetics*, **13**, 67–73.
- Matus J T, Aquea F, Arce-Johnson P. 2008. Analysis of the grape MYB R2R3 subfamily reveals expanded wine quality-related clades and conserved gene structure organization across *Vitis* and *Arabidopsis* genomes. *BMC Plant Biology*, **8**, 83.
- Mizoi J, Shinozaki K, Yamaguchi-Shinozaki K. 2012. AP2/ERF family transcription factors in plant abiotic stress responses. *Biochimica et Biophysica Acta* (BBA, Gene Regulatory Mechanisms), **1819**, 86–96.
- Muller D, Schmitz G, Theres K. 2006. *Blind* homologous R2R3 Myb genes control the pattern of lateral meristem initiation in *Arabidopsis*. *The Plant Cell*, **18**, 586–597.
- Nuruzzaman M, Manimekalai R, Sharoni A, Satoh K, Kondoh H, Ooka H, Kikuchi S. 2010. Genome-wide analysis of NAC transcription factor family in rice. *Gene*, **465**, 30–44.
- Ogata K, Morikawa S, Nakamura H, Sekikawa A, Inoue T, Kanai H, Sarai A, Ishii S, Nishimura Y. 1994. Solution structure of a specific DNA complex of the Myb DNA-binding domain with cooperative recognition helices. *Cell*, **79**, 639–648.
- Panchy N, Shiu M L, Shiu S H. 2016. Evolution of gene duplication in plants. *Plant Physiology*, **171**, 2294–2316.
- Passeri V, Koes R, Quattrocchio F M. 2016. New challenges for the design of high value plant products: Stabilization of anthocyanins in plant vacuoles. *Frontiers in Plant Science*, **7**, 9.
- Paz-Ares J, Ghosal D, Wienand U, Peterson P A, Saedler H. 1987. The regulatory c1 locus of *Zea mays* encodes a protein with homology to myb proto-oncogene products and with structural similarities to transcriptional activators. *EMBO Journal*, **6**, 3553–3558.
- Peel G J, Pang Y, Modolo L V, Dixon R A. 2009. The LAP1 MYB transcription factor orchestrates anthocyanidin biosynthesis and glycosylation in *Medicago*. *The Plant Journal*, **59**, 136–149.
- Perez-Diaz J R, Perez-Diaz J, Madrid-Espinoza J, Gonzalez-Villanueva E, Moreno Y, Ruiz-Lara S. 2016. New member of the R2R3-MYB transcription factors family in grapevine suppresses the anthocyanin accumulation in the flowers of transgenic tobacco. *Plant Molecular Biology*, **90**, 63–76.
- Perez-Rodriguez M, Jaffe F W, Butelli E, Glover B J, Martin C. 2005. Development of three different cell types is associated with the activity of a specific MYB transcription factor in the ventral petal of *Antirrhinum majus* flowers. *Development*, **132**, 359–370.
- Qin C, Cheng L, Shen J, Zhang Y, Cao H, Lu D, Shen C. 2016. Genome-wide identification and expression analysis of the 14-3-3 family genes in *Medicago truncatula*. *Frontiers in Plant Science*, **7**, 320.
- Riechmann J L, Heard J, Martin G, Reuber L, Jiang C Z, Keddie J, Adam L, Pineda O, Ratcliffe O J, Samaha R R, Creelman R, Pilgrim M, Broun P, Zhang J Z, Ghandehari D, Sherman B K, Yu G L. 2000. *Arabidopsis* transcription factors: Genome-wide comparative analysis among eukaryotes. *Science*, **290**, 2105–2110.
- Sagawa J M, Stanley L E, LaFountain A M, Frank H A, Liu C, Yuan Y. 2016. An R2R3-MYB transcription factor regulates carotenoid pigmentation in *Mimulus lewisii* flowers. *New Phytologist*, **209**, 1049–1057.
- Schmutz J, Cannon S B, Schlueter J, Ma J, Mitros T, Nelson W, Hyten D L, Song Q, Thelen J J, Cheng J, Xu D, Hellsten U, May G D, Yu Y, Sakurai T, Umezawa T, Bhattacharyya M K, Sandhu D, Valliyodan B, Lindquist E, et al. 2010. Genome sequence of the palaeopolyploid soybean. *Nature*, **463**, 178–183.
- Sharoni A M, Nuruzzaman M, Satoh K, Shimizu T, Kondoh H, Sasaya T, Choi I R, Omura T, Kikuchi S. 2011. Gene structures, classification and expression models of the AP2/

- EREBP transcription factor family in rice. *Plant and Cell Physiology*, **52**, 344–360.
- Shin J H, Van K, Kim D H, Kim K D, Jang Y E, Choi B S, Kim M Y, Lee S H. 2008. The lipoxygenase gene family: A genomic fossil of shared polyploidy between *Glycine max* and *Medicago truncatula*. *BMC Plant Biology*, **8**, 133.
- Shoemaker R C, Schlueter J, Doyle J J. 2006. polyploidy and gene duplication in soybean and other legumes. *Current Opinion in Plant Biology*, **9**, 104–109.
- Soler M, Camargo E L O, Carocha V, Cassan-Wang H, San Clemente H, Savelli B, Hefer C A, Paiva J A P, Myburg A A, Grima-Pettenati J. 2015. The *Eucalyptus grandis* R2R3-MYB transcription factor family: Evidence for woody growth-related evolution and function. *New Phytologist*, **206**, 1364–1377.
- Song H, Nan Z B. 2014. Genome-wide identification and analysis of WRKY transcription factors in *Medicago truncatula*. *Hereditas*, **36**, 152–168. (in Chinese)
- Song S S, Qi T C, Huang H, Ren Q C, Wu D W, Chang C Q, Peng W, Liu Y L, Peng J R, Xie D X. 2011. The jasmonate-ZIM domain proteins interact with the R2R3-MYB transcription factors MYB21 and MYB24 to affect jasmonate-regulated stamen development in *Arabidopsis*. *The Plant Cell*, **23**, 1000–1013.
- Stracke R, Holtgrawe D, Schneider J, Pucker B, Sorensen T R, Weisshaar B. 2014. Genome-wide identification and characterisation of R2R3-MYB genes in sugar beet (*Beta vulgaris*). *BMC Plant Biology*, **14**, 17.
- Stracke R, Werber M, Weisshaar B. 2001. The R2R3-MYB gene family in *Arabidopsis thaliana*. *Current Opinion in Plant Biology*, **4**, 447–456.
- Tamura K, Peterson D, Peterson N, Stecher G, Nei M, Kumar S. 2011. MEGA5: molecular evolutionary genetics analysis using maximum likelihood, evolutionary distance, and maximum parsimony methods. *Molecular Biology and Evolution*, **28**, 2731–2739.
- Uppalapati S R, Marek S M, Lee H-K, Nakashima J, Tang Y, Sledge M K, Dixon R A, Mysore K S. 2008. Global gene expression profiling during *Medicago truncatula*-*Phymatotrichopsis omnivora* interaction reveals a role for jasmonic acid, ethylene, and the flavonoid pathway in disease development. *Molecular Plant-Microbe Interactions*, **22**, 7–17.
- Walther D, Brunnemann R, Selbig J. 2007. The regulatory code for transcriptional response diversity and its relation to genome structural properties in *A. thaliana*. *PLoS Genetics*, **3**, 216–229.
- Wang H Y, Wang H L, Shao H B, Tang X L. 2016. Recent advances in utilizing transcription factors to improve plant abiotic stress tolerance by transgenic technology. *Frontiers in Plant Science*, **7**, 13.
- Wang W, Li E Y, Porth I, Chen J G, Mansfield S, Douglas C, Wang S C. 2016. Spatially and temporally restricted expression of *PtrMYB021* regulates secondary cell wall formation in *Arabidopsis*. *Journal of Plant Biology*, **59**, 16–23.
- Wilkins O, Nahal H, Foong J, Provart N J, Campbell M M. 2009. Expansion and diversification of the *Populus* R2R3-MYB family of transcription factors. *Plant Physiology*, **149**, 981–993.
- Xu G, Huang J, Yang Y, Yao Y A. 2016. Transcriptome analysis of flower sex differentiation in *Jatropha curcas* L. using RNA sequencing. *PLOS ONE*, **11**, doi: 10.1371/journal.pone.0145613
- Yang A, Dai X, Zhang W H. 2012. An R2R3-type MYB gene, *OsMYB2*, is involved in salt, cold, and dehydration tolerance in rice. *Journal of Experimental Botany*, **63**, 2541–2556.
- Young N D, Debelle F, Oldroyd G E D, Geurts R, Cannon S B, Udvardi M K, Benedito V A, Mayer K F X, Gouzy J, Schoof H, Van de Peer Y, Proost S, Cook D R, Meyers B C, Spannagl M, Cheung F, De Mita S, Krishnakumar V, Gundlach H, Zhou S G, et al. 2011. The *Medicago* genome provides insight into the evolution of rhizobial symbioses. *Nature*, **480**, 520–524.
- Yu Y T, Wu Z, Lu K, Bi C, Liang S, Wang X F, Zhang D P. 2016. Overexpression of the MYB37 transcription factor enhances abscisic acid sensitivity, and improves both drought tolerance and seed productivity in *Arabidopsis thaliana*. *Plant Molecular Biology*, **90**, 267–279.
- Zhang W, Han Z X, Guo Q L, Liu Y, Zheng Y X, Wu F L, Jin W B. 2014. Identification of maize long non-coding RNAs responsive to drought stress. *PLOS ONE*, **9**, doi: 10.1371/journal.pone.0098958
- Zhang Z Y, Liu X, Wang X D, Zhou M P, Zhou X Y, Ye X G, Wei X N. 2012. An R2R3 MYB transcription factor in wheat, TaPIMP1, mediates host resistance to *Bipolaris sorokiniana* and drought stresses through regulation of defense- and stress-related genes. *New Phytologist*, **196**, 1155–1170.
- Zheng K J, Tian H N, Hu Q N, Guo H Y, Yang L, Cai L, Wang X T, Liu B, Wang S C. 2016. Ectopic expression of R3 MYB transcription factor gene *OsTCL1* in *Arabidopsis*, but not rice, affects trichome and root hair formation. *Scientific Reports*, **6**, 12.
- Zhou M L, Hou H L, Zhu X M, Shao J R, Wu Y M, Tang Y X. 2011. Soybean transcription factor *GmMYB22* represses catharanthine biosynthesis in hairy roots of *Catharanthus roseus*. *Applied Microbiology and Biotechnology*, **91**, 1095–1105.

(Managing editor ZHANG Juan)